FNCE30007 DERIVATIVE SECURITIES
Week 7 Case Study - Virgin Australia

Case Background

Fuel price volatility is a major source of risk for airlines because jet fuel represents a significant share of operating costs. In the first half of FY2026, Virgin Australia spent A$554.7 million on fuel, equivalent to approximately 3.4 million barrels of oil and 21% of its total operating expenses. Virgin manages this exposure through a fuel-hedging program covering both Brent crude oil prices and jet-fuel refining margins.

This risk became particularly important in early 2026. Following major disruptions to global fuel markets, jet-fuel prices rose sharply and at one point more than doubled from their levels at the end of February. By April 2026, Virgin had hedged approximately 92% of its Brent crude oil exposure and 71% of its refining-margin exposure for the remainder of the financial year. The company estimated that the remaining unhedged exposure would increase its second-half fuel costs by approximately A$30–40 million.

The experience of Qantas illustrates why the structure of fuel hedging matters. Qantas had also hedged approximately 90% of its crude-oil exposure, but remained largely exposed to movements in jet-fuel refining margins. It subsequently revised its expected second-half fuel costs upward by approximately A$600–800 million. The comparison highlights the substantial financial consequences that changes in fuel prices can create for airlines.

Virgin uses derivatives, including options, as part of its fuel-risk management strategy. However, the detailed terms and premiums of its individual OTC contracts are not publicly available. In this case, we therefore use exchange-traded ICE Brent crude oil futures options as a transparent market proxy. Using real market prices, maturities and implied volatilities, we examine how accurately the Binomial Option Pricing Model can value these options under different model specifications.

All market data are as at 2 September 2026.

	Option A	**Option B**	**Option C**
Underlying futures contract	Nov 2026 Brent	Dec 2026 Brent	Jan 2027 Brent
Futures price, F_0	95.63	92.09	88.98
Call strike, K	95.00	92.00	89.00
Option expiry	25 Sep 2026	27 Oct 2026	25 Nov 2026
Days to expiry	23 (=0.063yr)	54 (=0.147yr)	83 (=0.227yr)
Implied volatility, σ	45.94%	46.88%	45.57%
Observed option price	4.67	6.70	7.72

The short-term Treasury rate (risk-free rate) is around r = 3.87% p.a.

1. Lecture Part: The Effect of Time to Maturity

1.1 Use a two-period binomial model to estimate the value of each of the three Brent call options, and complete the following table:

	Option A Nov 2026	**Option B Dec 2026**	**Option C Jan 2027**
Time to maturity	23 days	54 days	83 days
Two-period Binomial value			
Observed market price	4.67	6.70	7.72
Difference: Model − Market			
Absolute % pricing error			

Calculate: Absolute % Pricing Error = |Model Value − Market Price| / Market Price × 100

1.2 Compare the results across the three options. What pattern do you observe between time to maturity and the pricing error of the two-period binomial model? Briefly explain why a two-period tree may provide a relatively coarse approximation for a longer-dated option.

2. Tutorial Part: The Effect of Increasing the Number of Periods

Now focus on Option C: Jan 2027 Brent Call, which has the longest time to maturity.
Revalue the option using: N = 2, 4, 8, 16

2.1 For each value of N, recalculate $\Delta t = T/N$ and the corresponding u, d and p. Use the simulator to construct the larger binomial trees where appropriate and compare the estimates in this table.

Number of Periods, N	**Binomial Value**	**Observed Market Price**	**Difference: Model − Market**	**Absolute % Pricing Error**
2		7.72		
4		7.72		
8		7.72		
16		7.72		

2.2 Interpretation

- How does the estimated option value change as the number of periods increases?
- Does the binomial estimate appear to become more stable as the tree becomes finer?
- How does the pricing error change as N increases?
- Does increasing the number of periods necessarily make the model value exactly equal to the observed market price? Briefly explain.

Data note: Option-chain figures are based on Barchart market data observed on 2 September 2026. Prices are quoted in USD per barrel. Barchart reports an option point value of USD 1,000.

Suggested Answers

1.1 Reference Results

	Option A Nov 2026	Option B Dec 2026	Option C Jan 2027
Days to expiry	23	54	83
Two-period Binomial value	4.35	5.91	6.78
Observed market price	4.67	6.70	7.72
Difference: Model − Market	−0.32	−0.79	−0.94
Absolute % pricing error	6.84%	11.78%	12.13%

1.2 Suggested interpretation

The two-period binomial model produces the smallest pricing error for the shortest-maturity option. In this dataset, the pricing error becomes larger for the medium- and longer-maturity options.

One possible explanation is that a two-period tree provides only a coarse approximation of the futures price process. As time to maturity increases while the number of periods remains fixed, each individual time step becomes longer, so the tree represents fewer possible price paths over the life of the option.